

# Continuous-Time Kalman-Bucy Filtering for GPS with Nonlinear Extension

Ethan Lien

18 March 2026

## Abstract

This report documents the results of simulation of two continuous-time GPS receivers under a variety of geometric and initialization conditions. A set of common state dynamics and satellite observation SDEs are used in the implementation of the two filters. However, the Kalman-Bucy filter linearizes the observations while the Extended Kalman-Bucy filter utilizes true nonlinear observations. Simulation results demonstrate that both filters perform similarly to one another and, overall, demonstrate that well-placed satellite geometry results in robustness in estimate initialization sensitivity.

## 1 Dynamic state and observation model

Our objective is to derive a continuous-time estimator for a drone in 2D using measurements from four satellites. We define our state as

$$x_t = \begin{bmatrix} p_x(t) \\ p_y(t) \\ v_x(t) \\ v_y(t) \\ b(t) \end{bmatrix}$$

where  $p_t = (p_x, p_y)$  defines the position of the drone,  $v_t = (v_x, v_y)$  as the velocity of the drone, and  $b$  as the receiver clock bias. We use the following SDE to model the state dynamics subject to noise:

$$dx_t = Ax_t dt + G dW_t$$

where

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad G = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ \sigma_a & 0 & 0 \\ 0 & \sigma_a & 0 \\ 0 & 0 & \sigma_b \end{bmatrix}$$

with  $W_t \in \mathbb{R}^3$  a standard Brownian motions and noise parameters

$$\sigma_a = 0.5 \text{ m/s}^{3/2}, \quad \sigma_b = 0.1 \text{ m}/\sqrt{\text{s}}$$

For our satellite measurement, we define it through the following SDE:

$$dy_t = \underbrace{\begin{bmatrix} \|p_t - s_1\| + b_t \\ \|p_t - s_2\| + b_t \\ \|p_t - s_3\| + b_t \\ \|p_t - s_4\| + b_t \end{bmatrix}}_{h(x_t)} dt + \underbrace{\sigma_r I_4}_{R^{1/2}} dV_t$$

where  $V_t \in \mathbb{R}^4$  is a Brownian motion, the noise parameter is

$$\sigma_r = 5 \text{ meters}$$

and  $s_i$  are satellite positions defined by Table 1.

|       | Geometry   |               |
|-------|------------|---------------|
|       | Primary    | Degenerate    |
| $s_1$ | (20000,0)  | (20000,0)     |
| $s_2$ | (0,20000)  | (21000,2000)  |
| $s_3$ | (-20000,0) | (22000,-2000) |
| $s_4$ | (0,-20000) | (23000,1000)  |

**Table 1:** Satellite positions.

## 2 Estimator implementation

All numerical simulations operate with  $\Delta t = 0.01$  s across a time horizon of 60 seconds. For consistency, the true initial state and initial estimate covariance will be fixed to be:

$$x_0 = \begin{bmatrix} 0 & 0 & 10 & 2 & 50 \end{bmatrix}^T$$

$$S_0 = \text{diag}(100^2, 100^2, 20^2, 20^2, 200^2)$$

### 2.1 Linear Kalman-Bucy Filter

To implement the Linear Kalman-Bucy Filter (KBF), the observation SDE needs to be linearized. Electing to linearize around the current estimate  $\hat{x}_t$ , we have

$$h(x_t) \approx h(\hat{x}_t) + H(\hat{x}_t)(x_t - \hat{x}_t)$$

with each row in the Jacobian  $H(\hat{x}_t) \in \mathbb{R}^{4 \times 5}$  is defined as

$$H_i(\hat{x}_t) = \left[ \frac{p_x - s_{i,x}}{r_i} \quad \frac{p_y - s_{i,y}}{r_i} \quad 0 \quad 0 \quad 1 \right] \Big|_{p=\hat{p}_t}, \quad r_i = \|p - s_i\|_{p=\hat{p}_t}$$

To generate an estimate of the state, we wish to use the formulation we derived for linear time-varying systems. Our first step is to obtain an innovation process that isolates the new information the observation provides us. We can define it through the SDE

$$d\nu_t = dy_t - h(\hat{x}_t)dt$$

which, theoretically, isolates  $R^{1/2} dV_t$  from our measurement. The next component we need is estimate covariance, which we obtain by solving a matrix Riccati equation. The solution we are familiar with usually uses a matrix that operates linearly with the state in the deterministic part of the observation SDE. In our implementation, the best we can get is the Jacobian, which functions the same within our linearized  $h(x)$ . Thus, we have our matrix Riccati ODE

$$\dot{S}_t = AS_t + S_tA^T + GG^T - S_tH(\hat{x}_t)^T R^{-1}H(\hat{x}_t)S_t$$

Together with the innovation, we can use the following SDE for our state estimate dynamics:

$$d\hat{x}_t = A\hat{x}_t dt + S_tH(\hat{x}_t)^T R^{-1}d\nu_t$$

One final item I would like to iron out considers the discretization that needs to happen to simulate numerically. It appears that the state estimate requires the current observation, which itself requires the current state estimate. Though this has no issues in the continuous-time realm, putting this into code creates a circular argument. To resolve this, for a given state estimate, we can project it into the future deterministically via an Euler approximation:

$$M \triangleq \hat{x}_{\tau+1} = \hat{x}_\tau + A\hat{x}_\tau \Delta t$$

We can then use this for the next observation measurement, and then solve for its corresponding state estimate.

So, this will be how we implement the filter in code. We define relevant state vectors

$$\begin{aligned} x_{\tau+1} &= x_\tau + Ax_\tau \Delta t + G \Delta W_\tau \\ M_\tau &= \hat{x}_{\tau|\tau-1} = \hat{x}_{\tau-1} + A\hat{x}_{\tau-1} \Delta t \end{aligned}$$

We then solve for the innovation via

$$\begin{aligned} dy_\tau &= (h(M_\tau) + H(M_\tau)(x_\tau - M_\tau))\Delta t + R^{1/2} \Delta V_\tau \\ d\nu_\tau &= dy_\tau - h(M_\tau)\Delta t \end{aligned}$$

Next we obtain the covariance term  $S_\tau$  by solving the matrix Riccati ODE

$$\dot{S}_\tau = AS_\tau + S_\tau A^T + GG^T - S_\tau H(M_\tau)^T R^{-1} H(M_\tau) S_\tau$$

starting from an initial  $S_{\tau-1}$  through one time step  $\Delta t$ . Finally, we produce our estimate by

$$\hat{x}_{\tau+1} = \hat{x}_\tau + A\hat{x}_\tau \Delta t + S_\tau H(M_\tau)^T R^{-1} d\nu_\tau$$

The process needs to be initialized with initial state  $x_0$ , initial state estimate  $\hat{x}_0$ , and initial estimate covariance  $S_0$ . We also perform the above iteration with  $M_0 = \hat{x}_0$ .

## 2.2 Continuous Extended Kalman-Bucy Filter

Much of the implementation of the continuous extended Kalman-Bucy Filter (EKBF) is similar to the implementation of the KBF. The sole difference is the use of the nonlinear  $h(x_t)$  in the observation SDE. We will continue to use the Jacobian for a linearized approximation of  $S_t$  and use the same computation for the innovation  $d\nu_t$ . For full clarity, the simulation will use the following equations:

$$\begin{aligned} x_{\tau+1} &= x_\tau + Ax_\tau \Delta t + G \Delta W_\tau \\ M_\tau &= \hat{x}_{\tau|\tau-1} = \hat{x}_{\tau-1} + A\hat{x}_{\tau-1} \Delta t \\ dy_\tau &= h(x_\tau) \Delta t + R^{1/2} \Delta V_\tau \\ d\nu_\tau &= dy_\tau - h(M_\tau)\Delta t \\ \hat{x}_{\tau+1} &= \hat{x}_\tau + A\hat{x}_\tau \Delta t + S_\tau H(M_\tau)^T R^{-1} d\nu_\tau \end{aligned}$$

with  $S_\tau$  being the solution of the following matrix Riccati ODE from  $S_{\tau-1}$  through  $\Delta t$ :

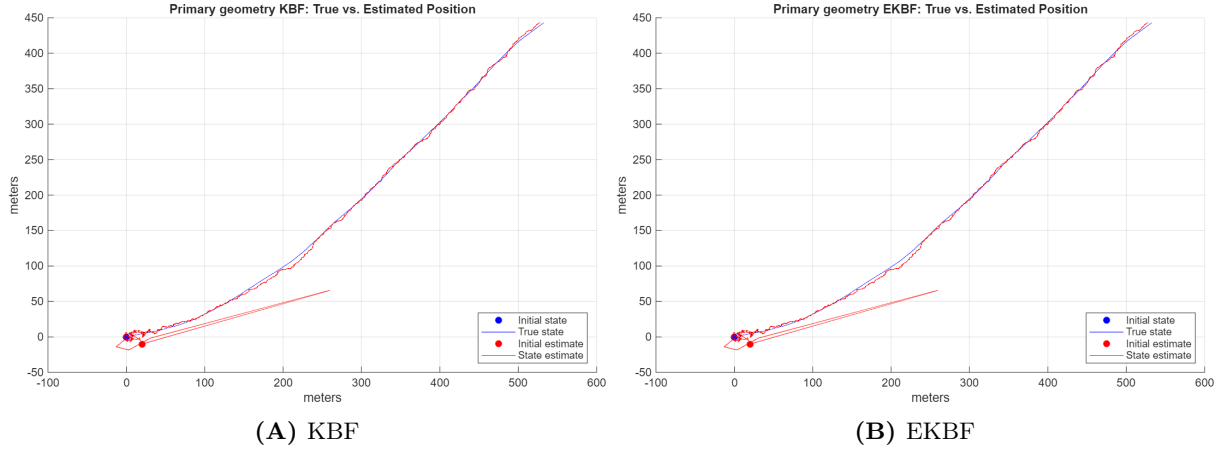
$$\dot{S}_\tau = AS_\tau + S_\tau A^T + GG^T - S_\tau H(M_\tau)^T R^{-1} H(M_\tau) S_\tau$$

## 3 Experiment 1: Baseline run

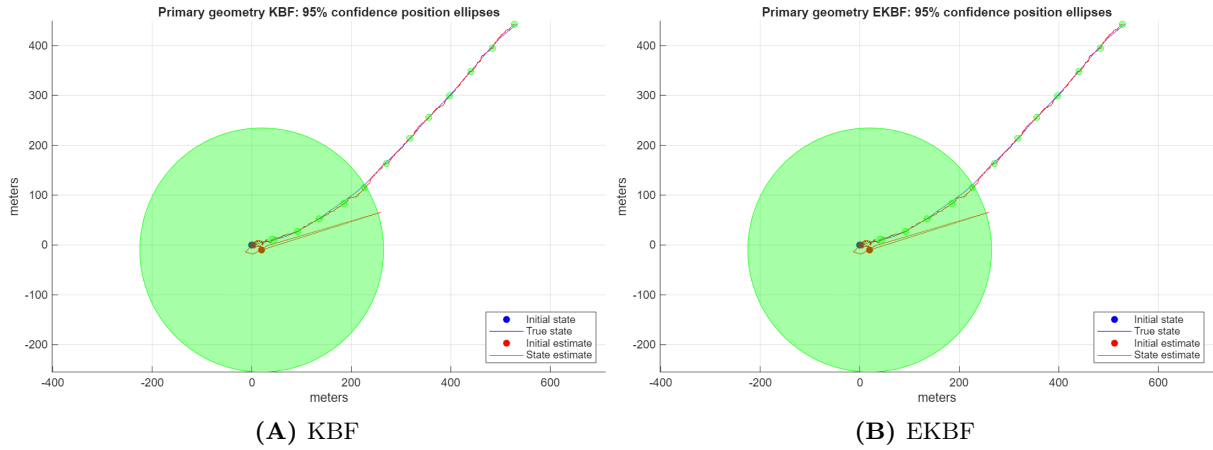
Experiment 1 functions as a control trial for the KBF and EKBF. These simulations will be using the primary (well-spread) satellite configuration and a nominal initial state estimate

$$\hat{x}_0 = \begin{bmatrix} 20 & -10 & 0 & 0 & 0 \end{bmatrix}^T$$

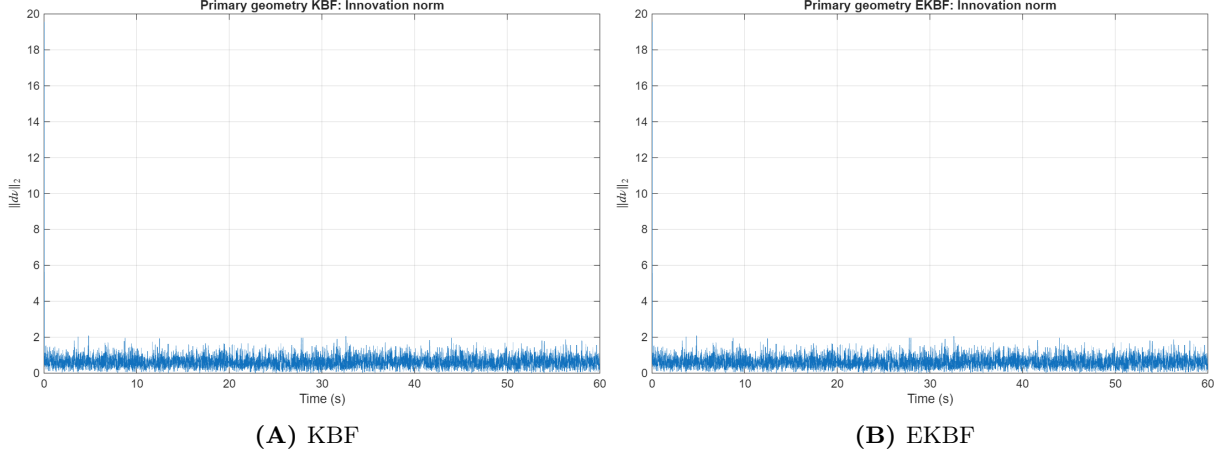
In the following simulations, both the KBF and EKBF are fed the same sequence of system disturbance and sensor noise.



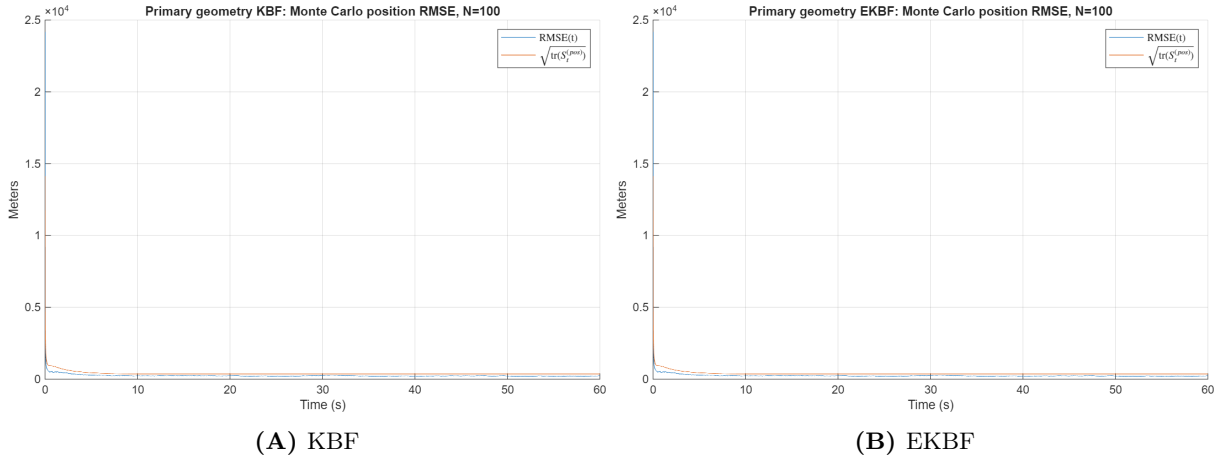
**Figure 1:** True and estimated state position trajectory. Primary geometry and nominal initial estimate.



**Figure 2:** Position 95% confidence ellipses. Primary geometry and nominal initial estimate.



**Figure 3:** Innovation norm over time. Primary geometry and nominal initial estimate.



**Figure 4:** Position RMSE and trace of estimated position covariance,  $N = 100$  Monte Carlo Simulations performed. Primary geometry and nominal initial estimate.

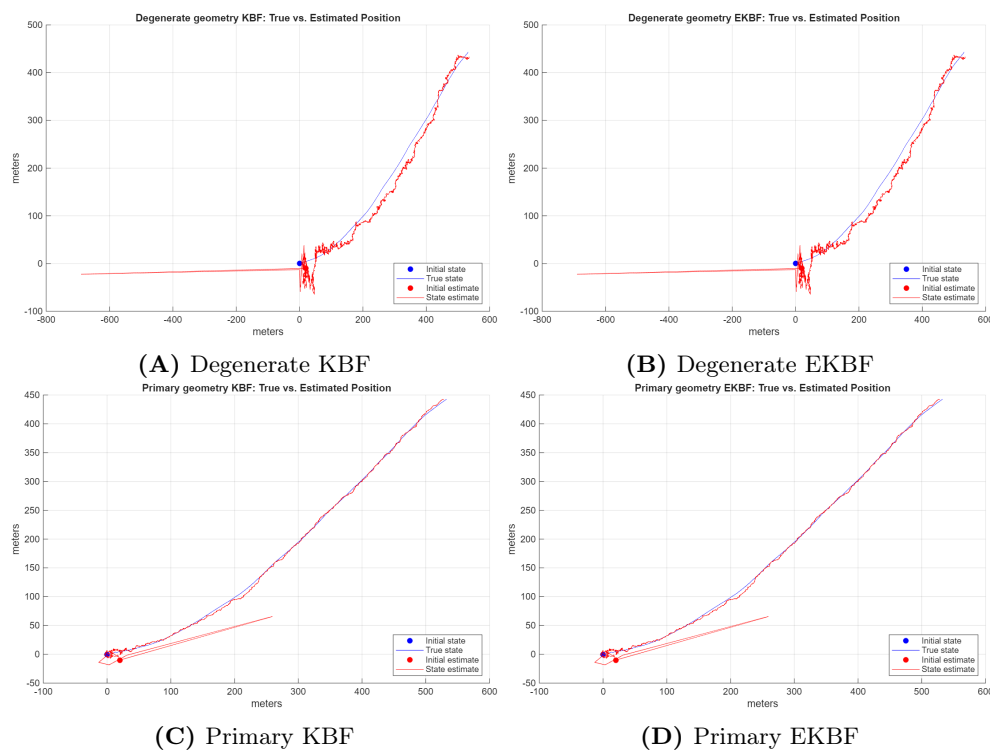
|      | $d\nu_1$          | $d\nu_2$          | $d\nu_3$          | $d\nu_4$          |
|------|-------------------|-------------------|-------------------|-------------------|
| KBF  | (-0.0052, 0.5298) | (-0.0027, 0.5344) | (-0.0032, 0.5479) | (-0.0135, 0.5469) |
| EKBF | (-0.0052, 0.5298) | (-0.0028, 0.5343) | (-0.0033, 0.5479) | (-0.0134, 0.5468) |

**Table 2:** Empirical  $d\nu_{t,i}$  distribution properties given by (mean,stdev).

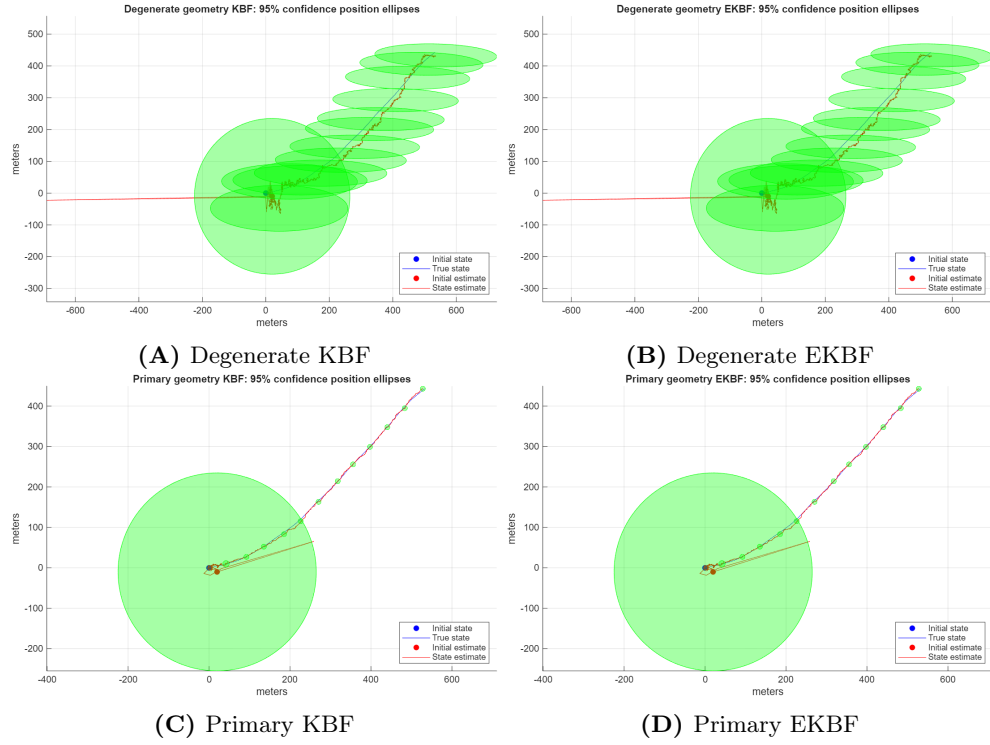
It appears that the KBF and EKBF performed near-identically. Both filters show great convergence properties, accurately estimating the position of the drone. They both show extreme (and accurate) confidence markers in both the  $x$  and  $y$  directions very soon after initialization. Moreover, the plots of the innovation norm indicate that new information is consistently fed into the estimator. It appears that the quantity  $\sqrt{\text{tr}(S_t^{(pos)})}$  is a consistent bound for the root mean squared error (RMSE) of the position estimate, meaning that our covariance is overly conservative and believes that its estimates are worse than they actually are.

## 4 Experiment 2: Degenerate geometry study

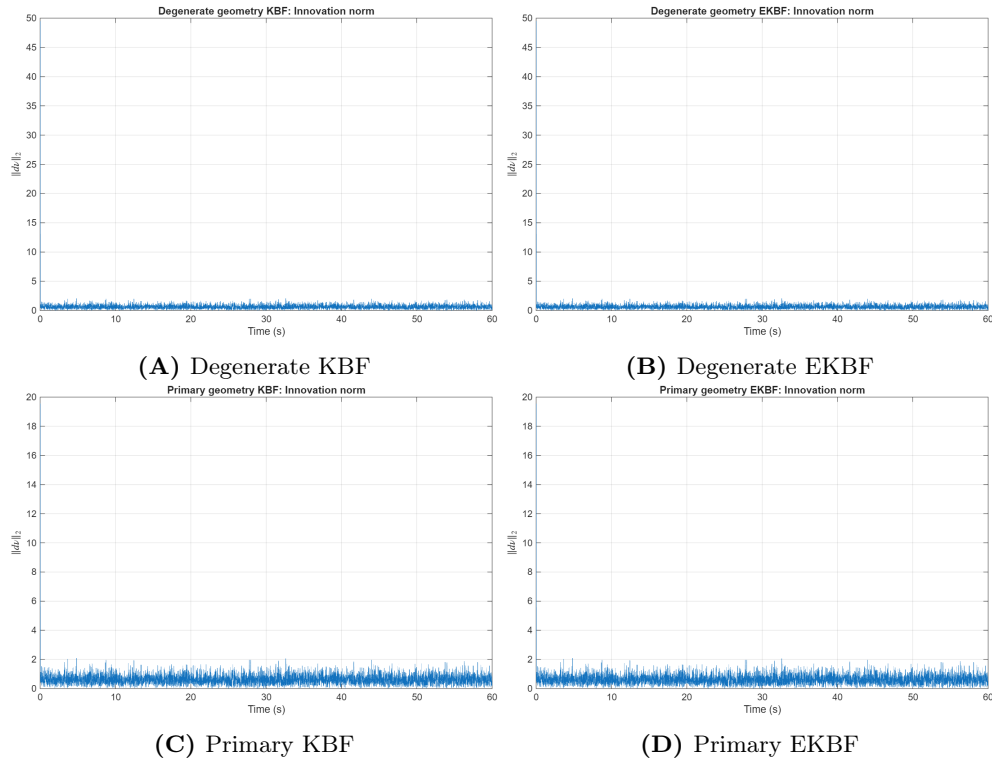
This experiment focuses on the effects of using clustered (degenerate) satellite geometry. The same nominal initial estimate is used for this study.



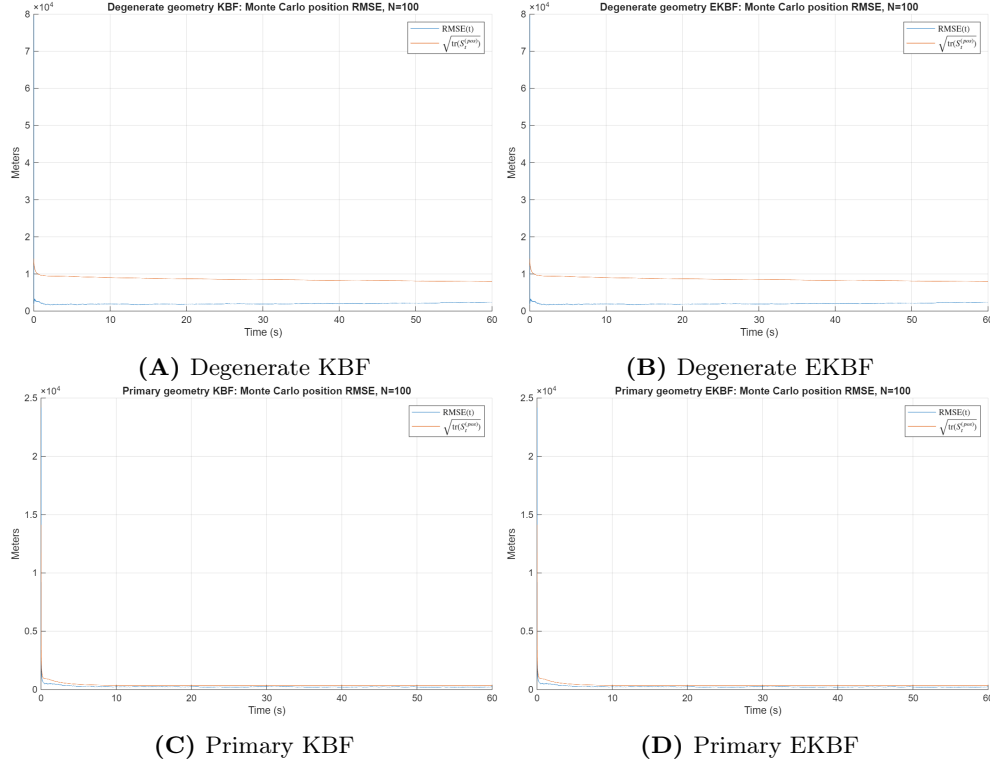
**Figure 5:** True and estimated state position trajectory. Nominal initial estimate.



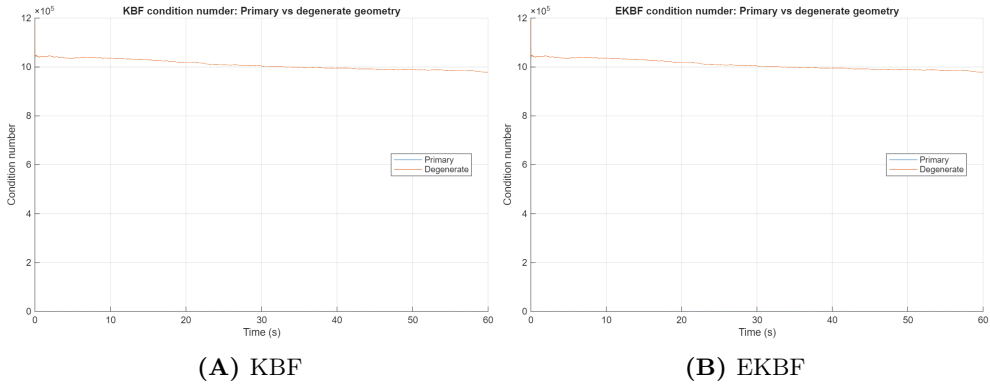
**Figure 6:** Position 95% confidence ellipses. Nominal initial estimate.



**Figure 7:** Innovation norm over time. Nominal initial estimate.



**Figure 8:** Position RMSE and trace of estimated position covariance,  $N = 100$  Monte Carlo Simulations performed. Nominal initial estimate.



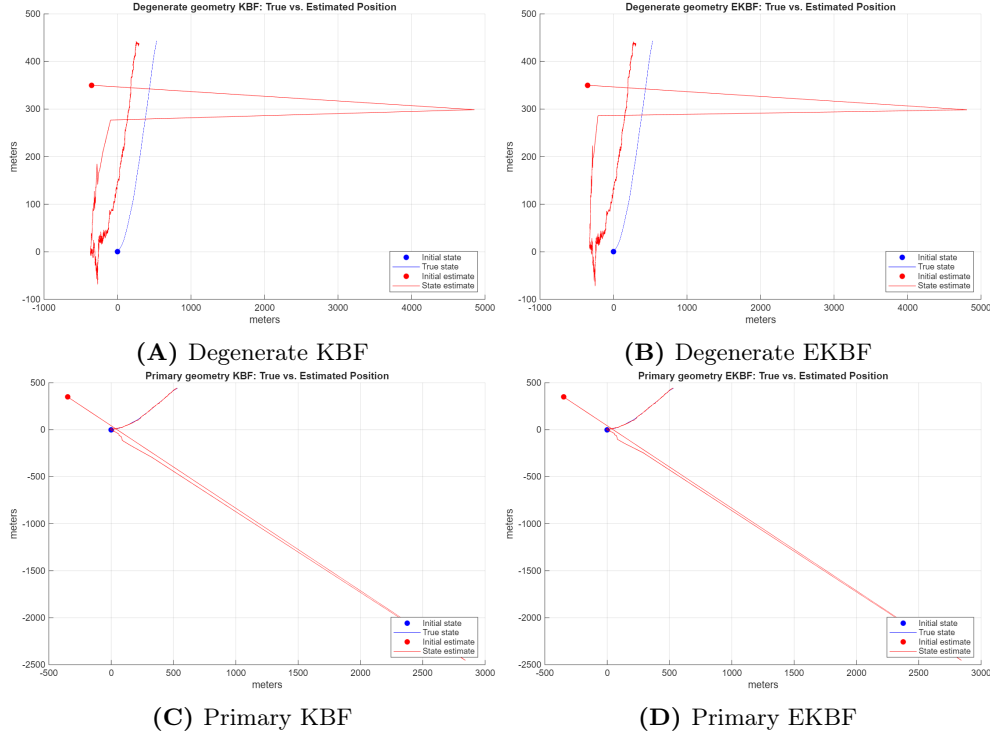
**Figure 9:** Condition number of  $H_t^T R^{-1} H_t$ , with zero columns of  $H_t$  ignored. Nominal initial estimate.

These results are pretty insightful. From Figure 5 that the convergence properties of the degenerate satellite geometry are worse — it takes longer for the estimated state to converge to the true state and there is a larger error between the true and estimated position. From the confidence ellipses, we can tell that there is great uncertainty in the estimate (large area), and that the confidence is asymmetric with respect to the  $x$  and  $y$  directions. This is due to the fact that this spread of satellites can report a mediocre variety of information along the  $y$ -direction but receive practically the same information along the  $x$ -direction. This is also demonstrated in Figure 7, which reports a smaller innovation norm for the degenerate case and reflects a reduction in new information entering the estimator, as well as in Figure ??, whose large condition number for the degenerate case reflects an incredibly large geometric dilution of precision. Finally we see that the internal model for estimate covariance are remarkably higher in the degenerate case, which less accurately reflect the empirical performance of the estimator.

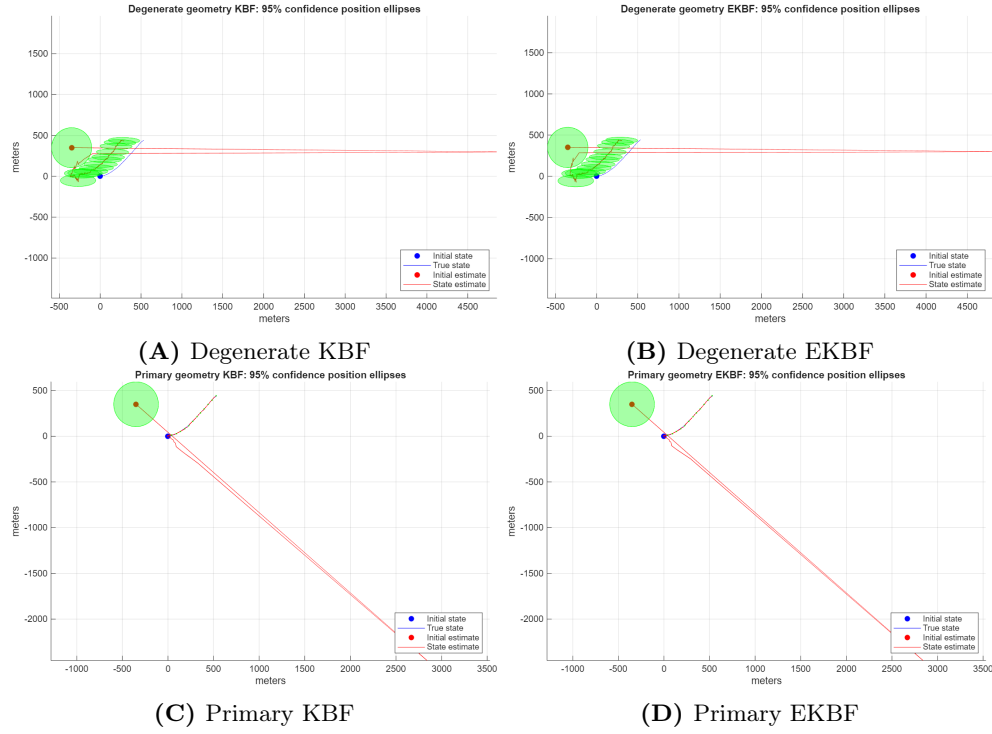
## 5 Experiment 3: Initialization sensitivity

The following experiment expands upon the previous experiments, examining both primary and degenerate geometries. However, for these simulations the following erroneous initial estimate will be used:

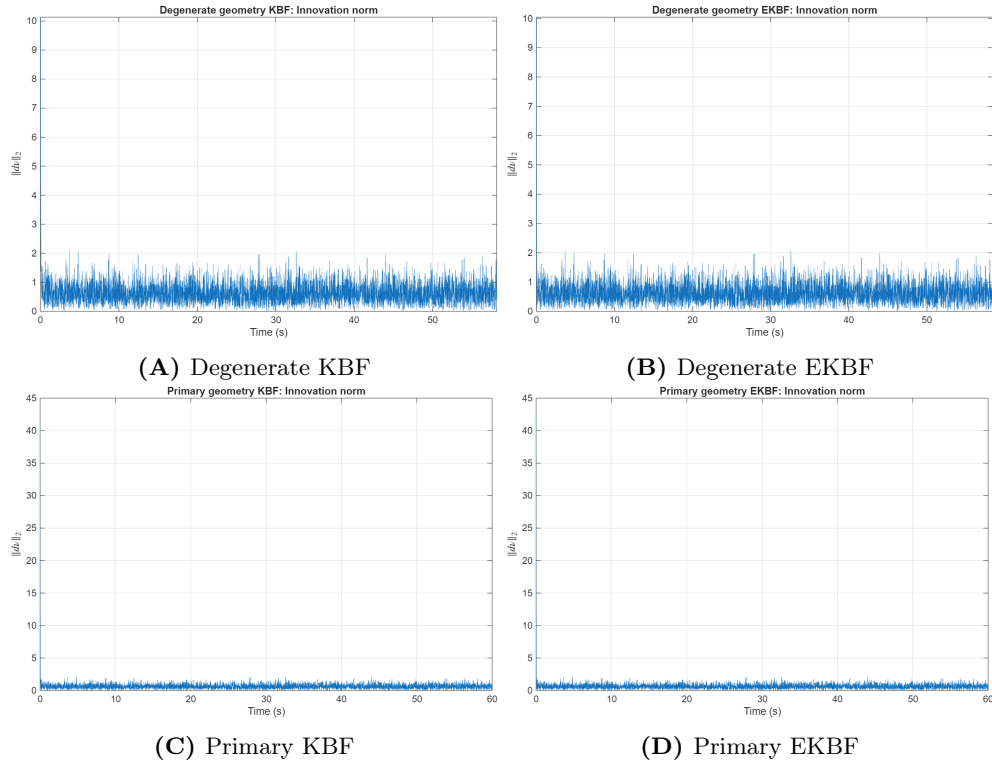
$$\hat{x}_0 = \begin{bmatrix} -350 & 350 & 0 & 0 & 0 \end{bmatrix}^T$$



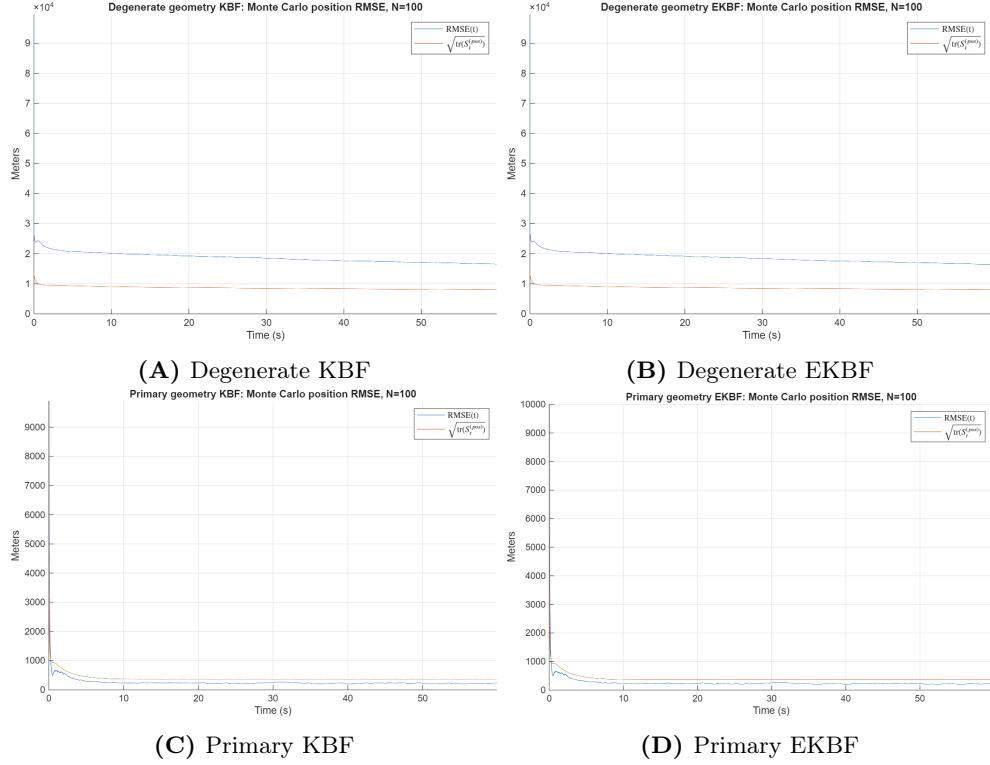
**Figure 10:** True and estimated state position trajectory. Erroneous initial estimate.



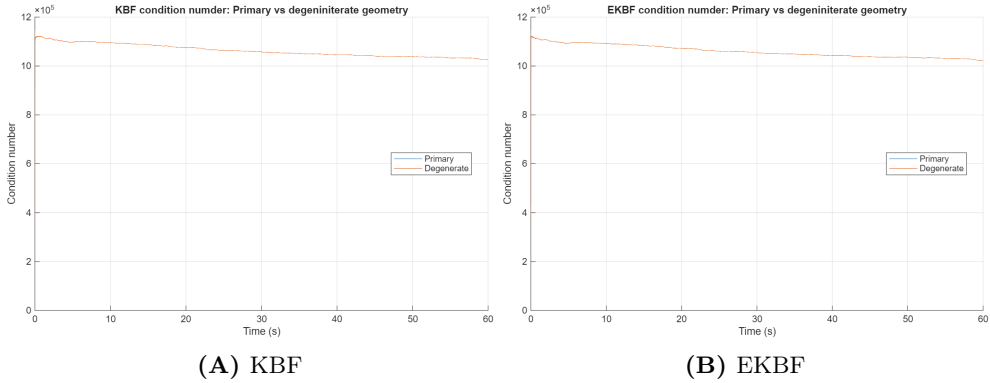
**Figure 11:** Position 95% confidence ellipses. Erroneous initial estimate.



**Figure 12:** Innovation norm over time. Erroneous initial estimate.



**Figure 13:** Position RMSE and trace of estimated position covariance,  $N = 100$  Monte Carlo Simulations performed. Erroneous initial estimate.



**Figure 14:** Condition number of  $H_t^T R^{-1} H_t$ , with zero columns of  $H_t$  ignored. Erroneous initial estimate.

It seems that primary geometry convergence is still guaranteed with an erroneous initial estimate. It seems that sufficient variety of information is able to be provided to our estimator, allowing for accurate estimation of drone position. We see the same position confidence, innovation, and RMSE trends as the nominal initial estimate cases.

The degenerate geometry, however, presents very different results. We can see that the estimated state does not converge to the true state. To be specific, the  $x$ -position of the estimated state does not converge to the true position while the  $y$ -position *does* converge. This is somewhat expected, as it is an extension of the geometric properties discussed in Experiment 2, where we have rich information along the  $y$ -direction and poor information along the  $x$ -direction. This is again supported by the confidence ellipses and the condition number evolution, which show a severe imbalance in confidence between the  $x$  and  $y$  directions.

What is new and interesting is the difference in position RMSE plots. It is showing that estimator is generally over-confident — that the position error is greater than what it is predicting/modeling.

## 6 Conceptual Questions

### Question 1

Without considering clock bias, it makes sense that a 2D drone would need 2 satellites to properly estimate its state. It can be considered that each satellite "handles" a single degree of freedom of the system ( $x$  and  $y$  position). When an unknown clock bias is present, it would be wonderful to know/estimate what it is so we can account for it in our internal system. The way we have done this was by introducing another state into our state vector. This introduces another degree of freedom that cannot be accounted for by 2 satellites alone. This can be further elaborated by considering observability. We can consider our system observable if

$$\text{rank} \left( \begin{bmatrix} h(x_t) \\ h(x_t)A \\ h(x_t)A^2 \\ \vdots \\ h(x_t)A^{n-1} \end{bmatrix} \right) = n$$

In our case, we can only look at

$$\text{rank} \left( \begin{bmatrix} h(x_t) \\ h(x_t)A \end{bmatrix} \right) \stackrel{?}{=} n$$

since  $A^p = 0$ ,  $p \geq 2$ . We can see clearly that in the case that we have only 2 satellites that the maximum possible rank that the observability matrix is 4, since  $h(x_t)$  would only have 2 rows. This is less than the 5 states our system has, so the observability condition is unable to be achieved. However, with the addition of a third satellite, it is possible for the rank condition to be met, meaning that position and clock bias can be accurately estimated.

### Question 2

Satellite geometry greatly affects our estimators' confidence in state estimate. The results presented in Figure 8 demonstrate that clustered satellite geometry blows up the estimation covariance matrix. This can be related to GDOP in the system. Because the satellites are so close to one another, there is a relatively low amount of new position information compared to sensor noise entering the estimator. As such, the condition number of  $H_t^{(g)T} H_t^{(g)}$  explodes and the GDOP vastly increases, which means that measurement noise has a greater impact in our observation process and position uncertainty.

### Question 3

From our experiments, it appears that the EKF diverges under erroneous initial state estimate and degenerate geometry. Whenever well-spread satellite geometry is used, the information entering the measurement system is mostly rich enough to correct for any error accumulated by the system (see Figure 5). Moreover, the degenerate satellite condition performs well enough so long as the initial state error is small (see Figure 10). We can see that the only case where there is no convergence of the EKF is in Figure 10(B). This situation falls prey to bad initial estimate which cannot be corrected due to the lack of position confidence from degenerate geometry.

## Question 4

|      | $d\nu_1$          | $d\nu_2$          | $d\nu_3$          | $d\nu_4$          |
|------|-------------------|-------------------|-------------------|-------------------|
| KBF  | (-0.0052, 0.5298) | (-0.0027, 0.5344) | (-0.0032, 0.5479) | (-0.0135, 0.5469) |
| EKBF | (-0.0052, 0.5298) | (-0.0028, 0.5343) | (-0.0033, 0.5479) | (-0.0134, 0.5468) |

**Table 3:** Empirical  $d\nu_{t,i}$  distribution properties given by (mean, stdev). Nominal initial estimate and primary satellite geometry.

The innovation process does not behave like Brownian motion. The above tables present the statistical distribution properties of the innovation process  $d\nu_t$ . We can see that they are zero-mean, which does match with Brownian motion. However, the innovation process does not have the expected standard deviation. For time increments of  $\Delta t = 0.01$  s, we expect a Brownian motion to have standard deviation of  $\sqrt{\Delta t} = 0.1$ . However, even in the best case scenario (nominal initial estimate and primary geometry) the standard deviation is around 0.5. This is due to the sensor noise that enters the system, which is a Brownian motion that is scaled by  $R^{1/2} = \sigma_r I_4$  in the observation SDE. When we look at the KBF innovation SDE,

$$\begin{aligned}
d\nu_t &= dy_t - h(\hat{x}_t)dt \\
&= H(\hat{x}_t)(x_t - \hat{x}_t)dt + \sigma_r dV_t \\
\mathbb{E}[d\nu_t d\nu_t^T] &= \mathbb{E}[H(\hat{x}_t) \underbrace{(x_t - \hat{x}_t)(x_t - \hat{x}_t)^T}_{S_t} H(\hat{x}_t)^T dt^2 + \underbrace{[\text{cross terms}]}_{\rightarrow 0, \text{ by indep. + zero-mean}} + \sigma_r^2 dV_t^2] \\
&= H(\hat{x}_t) S_t H(\hat{x}_t)^T dt^2 + \sigma_r^2 \Delta t
\end{aligned}$$

We see that the variance of this is almost certainly never going to be  $\Delta t$ . Thus, this is not a Brownian motion. Another check could be for the independence of the increments. Assuming that they are Gaussian in distribution, we can check the covariance — if the innovation is independent, then the covariance matrix should be near diagonal<sup>1</sup>. Checking empirically, we get the following for one trial of the experiment:

$$\text{covariance} = \begin{bmatrix} 0.2807 & 0.0300 & 0.0334 & 0.0454 \\ 0.0300 & 0.2855 & 0.0445 & 0.0403 \\ 0.0334 & 0.0445 & 0.3002 & 0.0464 \\ 0.0454 & 0.0403 & 0.0464 & 0.2990 \end{bmatrix}$$

This is very much not diagonal, thus,  $d\nu_t$  is not a Brownian motion.

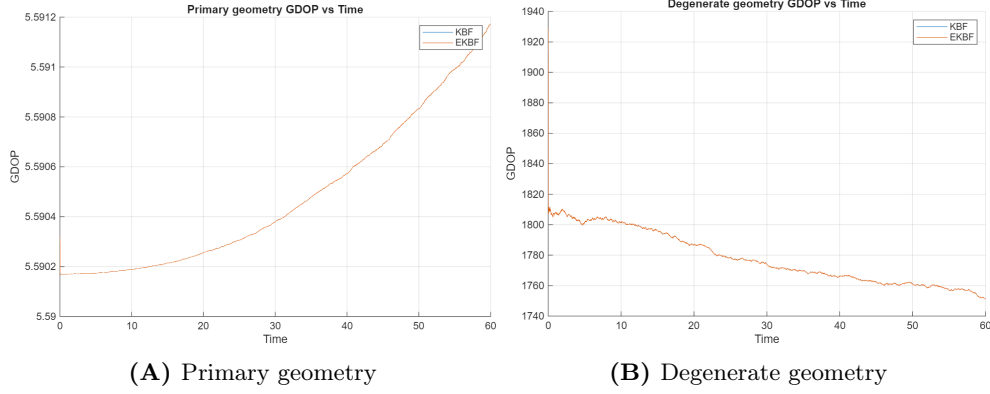
## 7 Conclusion

We have successfully implemented two variations of continuous-time GPS receiver. We have seen that well-spread satellite geometry can successfully compensate for poor initial state estimates, reducing estimation error and providing great, isotropic estimate confidence. However, clustered satellite geometry is more sensitive to initial state estimation, as less information is obtained by the estimator, leading to larger directional estimate errors and even larger estimation covariance values. Overall, it appears that the KBF and the EKBF performed similarly to one another when provided the same experimental conditions.

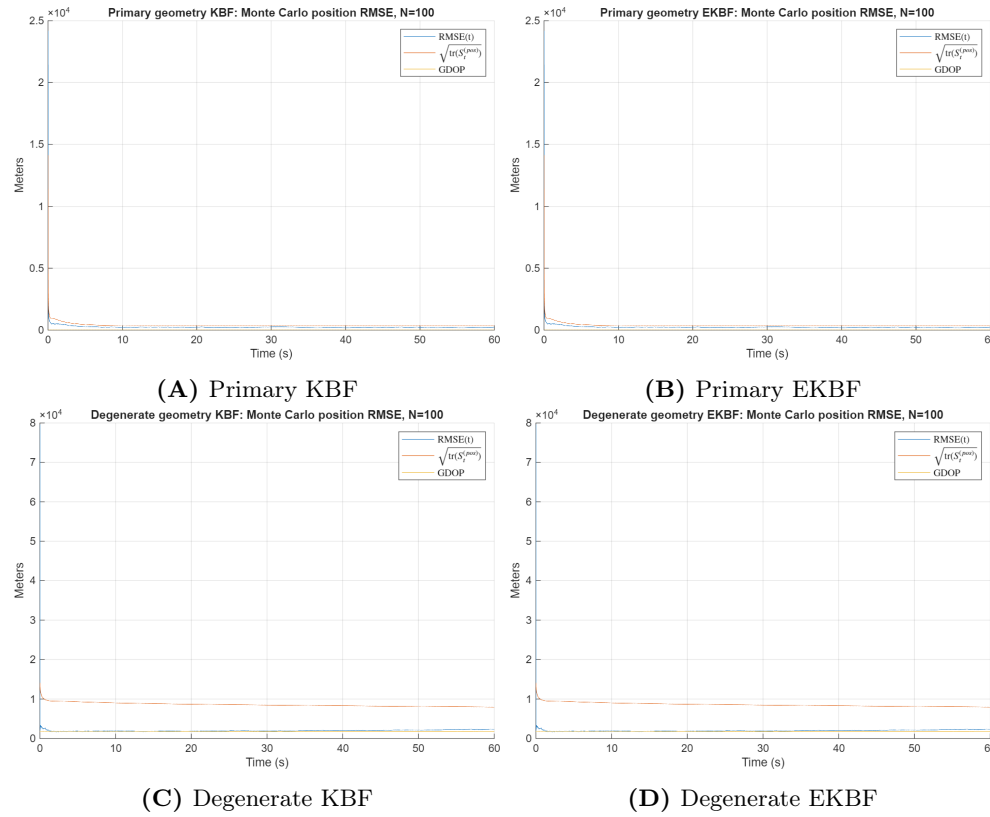
<sup>1</sup> This actually checks for correlation, but with a Gaussian distribution, is the same as independence.

## 8 Extra Analysis

### 8.1 Geometric Dilution of Precision



**Figure 15:** Evolution of system GDOP over time. Nominal initial estimate.

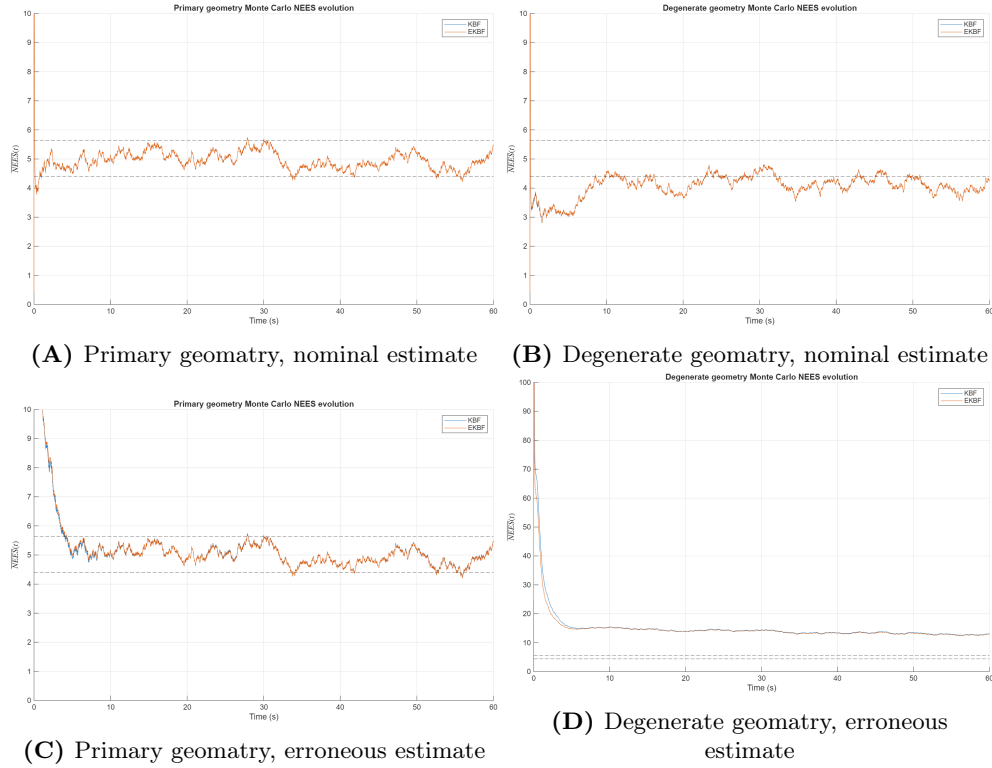


**Figure 16:** Position RMSE trace of estimated position covariance,  $N = 100$  Monte Carlo Simulations performed, with trial GDOP. Nominal initial estimate.

We can see that the simulation results align with our intuition. We see that with the well-spread geometry that the GDOP is very low and consistent throughout the simulation, signifying that the measurement noise is less amplified in our position uncertainty. On the other hand, with the clustered geometry the GDOP values explode, meaning noise has a greater impact in our position confidence. We can see in the degenerate

case that the GDOP and empirical RMSE seem to converge, with the GDOP almost acting as a lower bound for the RMSE. This means that we the estimate error in the degenerate case will consistent have a large covariance and will experience much slower convergence to the true state.

## 8.2 Monte Carlo Consistency Analysis



**Figure 17:** Monte Carlo Normalized Estimation Error Squared evolution over time,  $N = 100$ . Consistency interval denoted by dashed black lines.

It appears that the KBF and EKBF perform very similarly for all configurations, though the KBF slightly outperforms in the erroneous initial estimate conditions. We see that for the primary geometry the filters are both statistically consistent and are accurately confident. The degenerate geometry cases are a bit odd. With a nominal initial estimate, the filters are mildly over-conservative, meaning their covariances are larger compared to their actual empirical RMSE. On the other hand, with an erroneous initial estimate the filters are incredibly overconfident. The empirical covariance is much higher than what the filters internally believes.

## 8.3 Summary Discussion

### GDOP and Riccati

We can see that the GDOP and Riccati contraction rate are directly related. Poor satellite geometry results in larger GDOP and greater amplification of sensor noise into the covariance of the the estimate. As a result, the integration of the Riccati equation is slower and settles to a larger steady state value.

### Consistency properties

From the results I obtained indicate that the KBF and EKBF perform similarly. The consistency depends on the satellite geometry and initialization conditions, for which primary geometry is the most accurate

and degenerate geometry with erroneous initial estimate performs the worst. These results align with the simulations results from the main experiments.